

Supervised Learning Report Rubric

CS7641: Machine Learning

Summer 2026

Schedule

- **6/15/2026 (Monday):** SL Grading Begins & Overview of Rubric
- **6/17/2026 (Wednesday):** SL First 2 Calibration
- **6/19/2026 (Friday):** SL Calibration with Examples and Head TAs Check
- **6/23/2026 (Tuesday):** SL 33–50% Completed
- **6/26/2026 (Friday):** SL 50–80% Completed
- **6/30/2026 (Tuesday):** SL 100% Completed
- **7/1/2026 (Wednesday):** SL Posted

Total: 100 points + 5 extra credit

This rubric is aligned to the Summer 2026 assignment specification and FAQ.

How to Use This Rubric

This rubric is designed to support **consistent grading across TAs** while still rewarding strong scientific work.

Each line item has a **fixed point value**. This is an all-or-nothing rubric. Either the requirement is clearly met and the points are awarded, or it is not met and the points are not awarded. Do not award partial credit within a listed item unless that item explicitly separates points into smaller sub-items.

Award points **only** when the evidence is explicit, correct, and directly supported by the report. If the report suggests a requirement was met but does not clearly demonstrate it, treat the item as not satisfied.

The reproducibility sheet may contain extended EDA, environment details, and implementation details. However, the main report must still stand alone as a formal technical report. Students should not move all substantive interpretation into the reproducibility sheet.

Minimum Standard for Hyperparameter Exploration

When a rubric item requires “multiple values,” the default expectation is at least **three distinct and reasonable values** for the relevant scalar hyperparameter unless the student gives a clear computational justification for a smaller but still meaningful sweep.

A reasonable sweep should cover a range broad enough to reveal underfitting, overfitting, or performance saturation. Testing tiny variations around one value, such as $k = 5, 6, 7$, does not count as meaningful exploration unless justified by prior evidence.

For computationally expensive models, especially SVMs, a smaller sweep may be acceptable if the student explains the computational constraint, reports runtime evidence, and still compares enough settings to support the final model choice.

Required Task

Students must complete the required task exactly as follows:

- **Dataset:** Forest Cover Type / Covertype
- **Task:** Multiclass classification
- **Target:** `Cover_Type`, classes 1 through 7
- **Source:** UCI Machine Learning Repository, Canvas, or Python import using a standard library such as `sklearn.datasets.fetch_covtype`
- **Sample size:** Approximately 20,000 instances sampled from the full dataset
- **Required metrics:** Accuracy, Macro-F1, balanced accuracy, confusion matrix, and per-class precision, recall, and F1 discussion

Sampling Expectation

The full Covertype dataset contains more than half a million instances. Students are required to create and justify a smaller working dataset of approximately **20,000 instances**.

A **stratified sample is the default expectation** because the target classes are imbalanced. A simple random sample may be acceptable only if the student demonstrates that all classes are preserved and that the sampled class distribution approximately reflects the full class distribution.

Students should **not** use the first 20,000 rows unless they provide a compelling justification. The sampling procedure must avoid accidental ordering effects.

Required Models

Students must run all four algorithm families on the sampled Covertype dataset. The goal is not for every student to reach the same final ranking of models. The goal is for the student to run the required algorithm families in a reproducible way, tune them with enough evidence to support claims, and explain model behavior using the dataset, curves, metrics, runtime, and class-level errors.

Covtype is a mixed-feature, multiclass, imbalanced tabular dataset. It contains continuous cartographic variables along with binary wilderness-area and soil-type indicators. This structure should shape model behavior. Distance-based and margin-based models should be sensitive to scaling. Tree-based models may exploit threshold-style splits without scaling. Neural networks may be sensitive to learning rate, initialization, batch size, scaling, and the SGD-only constraint. TAs should not expect one universal ranking of algorithms, but they should expect students to explain their observed ranking using evidence.

For all model families, students should report final metrics using Accuracy, Macro-F1, and balanced accuracy, and should discuss class-level behavior using a confusion matrix or per-class precision/recall/F1. Strong reports will notice when high Accuracy hides poor minority-class performance. Weak reports will often declare a winner based only on Accuracy.

Decision Trees

Decision Trees serve as an interpretable nonlinear baseline. Students must control complexity using at least one pruning or regularization mechanism, such as `max_depth`, `min_samples_leaf`, `min_samples_split`, or cost-complexity pruning with `ccp_alpha`. Gini and entropy are both acceptable splitting criteria as long as the student states what was used.

TAs should expect a Decision Tree model-complexity curve over a tree-specific capacity parameter. A useful curve might vary depth, minimum leaf size, or `ccp_alpha`. The interpretation should explain how the chosen setting balances underfitting and overfitting. A shallow tree may underfit if both training and validation scores are low. An unconstrained or very deep tree may show high training performance with lower validation performance, which is evidence of variance or overfitting. A strong report will connect this behavior to the learning curve or pruning curve.

Plausible Covertypes behavior: Decision Trees may perform reasonably well because the data include threshold-friendly continuous variables and binary indicators. However, a single tree may be less stable than more flexible methods and may overfit minority or local patterns if not regularized. Strong students may note that trees are relatively insensitive to monotonic feature scaling, making them a useful comparison point against kNN, SVM, and neural networks.

Common weak submissions: no pruning or regularization; only one tree setting; reporting final depth without explaining why it was selected; showing a complexity curve but not connecting it to bias/variance; reporting high Accuracy without discussing minority-class recall.

k-Nearest Neighbors

kNN is a locality-based method. Students must use appropriate scaling, state the distance metric, state whether uniform or distance weighting was used, and explore meaningfully different values of k . At least three distinct k values are expected unless a computational justification is provided. Values should represent different neighborhood sizes rather than tiny variations around one setting.

TAs should expect the kNN model-complexity curve to show validation performance as k changes. Small k values often have lower bias but higher variance, while large k values smooth local structure and may hurt minority classes if those classes are sparse. The student should explain whether the selected k appears to balance local sensitivity and generalization. The learning curve may show whether kNN benefits from more training data, but students should also discuss prediction-time cost, since kNN often has cheap fitting but expensive prediction.

Plausible Covertypes behavior: kNN may perform competitively if nearby points in scaled feature space tend to share cover types. It may struggle when classes overlap or when minority classes have sparse local neighborhoods. The mixed continuous and binary feature structure makes scaling and distance interpretation important. If continuous variables are not scaled, large-range features may dominate the distance calculation and produce misleading neighborhoods.

Common weak submissions: no scaling; scaling fit before the train/test split; only testing one k ; testing values such as $k = 5, 6, 7$ without justification; reporting kNN as “fast” because fit time is small while ignoring prediction time; failing to discuss how k affects minority-class behavior.

Support Vector Machines

SVMs are margin-based models and should be evaluated with at least two kernels, such as linear and RBF. Students must scale features in a leakage-safe way. Students should tune C , and for RBF or other nonlinear kernels they should also tune the relevant kernel parameter such as γ . For computationally expensive SVM experiments, a reduced validation strategy or smaller tuning grid is acceptable only if the student explains the runtime constraint and still provides enough evidence to justify the comparison.

TAs should expect students to compare kernel behavior, not merely report that two kernels were run. A linear SVM tests whether a linear decision boundary in the scaled feature space is sufficient. An RBF SVM tests whether nonlinear boundaries improve class separation. The model-complexity curve may vary C , γ , or one of them while holding the other fixed. A strong explanation will connect C and γ to margin flexibility, overfitting, underfitting, and runtime.

Plausible Covertypes behavior: an RBF SVM may outperform a linear SVM if cover types are not linearly separable in the original feature space. However, RBF SVMs may be expensive on 20,000 instances, especially with cross-validation and broad grids. A student who uses a smaller validation split, a reduced grid, or a staged search can still be scientifically reasonable if they report the constraint clearly and use the evidence honestly. A linear SVM may be faster and more stable but may underfit if class boundaries are nonlinear.

Common weak submissions: no scaling; only one kernel; mentioning RBF but not tuning γ ; choosing the final SVM based on test performance; using an extremely small or arbitrary subset without justification; reporting that SVM was “too slow” without runtime evidence; failing to compare kernel trade-offs.

Neural Networks

Neural Networks must be trained under the SGD-only constraint and must include two implementations:

- **scikit-learn:** `MLPClassifier` trained with SGD.

- **PyTorch:** a compact MLP trained with SGD.

The SGD constraint is strict. Students may not use momentum, Nesterov momentum, Adam, AdamW, AdaGrad, RMSprop, or other adaptive optimizers. For scikit-learn, this means `solver='sgd'`, `momentum=0`, and `nesterovs_momentum=False`. For PyTorch, this means plain `torch.optim.SGD` without momentum.

TAs should expect students to document learning rate, batch size, number of epochs, early stopping if used, architecture, activation, regularization, and random seed where applicable. Students should show epoch-based training and validation curves for both implementations. These curves are important because the neural-network requirement is not only about final performance; it is also about optimization behavior under plain SGD.

Plausible Covertypes behavior: neural networks may perform well if scaling and learning rate are handled carefully, but plain SGD can be unstable or slow compared with adaptive optimizers. It is plausible for neural networks to underperform stronger SVM or kNN configurations if learning rate, initialization, architecture, or training budget are not well controlled. It is also plausible for PyTorch and scikit-learn to differ because of defaults, initialization, batching, stopping criteria, implementation flexibility, and training-loop control.

Strong reports will compare the two implementations in terms of optimization stability, epoch curves, runtime, tunability, final metrics, and class-level behavior. They should not simply report two final scores. The required comparison should address whether one implementation was easier to stabilize, whether one showed smoother convergence, whether either appeared undertrained or overfit, and whether runtime or implementation flexibility affected the final result.

Common weak submissions: using Adam or momentum; failing to state optimizer settings; showing only final NN metrics without epoch curves; reporting only one implementation; comparing scikit-learn and PyTorch as if they should be numerically identical; changing many architecture and optimization settings at once without interpretation; ignoring instability or divergence in the epoch curves.

General TA Expectations Across Models

Students do not need exhaustive hyperparameter searches. A full grid search is not required and may be a poor use of computation. However, each model family should have enough exploration to support the final choice. In most cases, this means at least three reasonable values for a key scalar hyperparameter and a model-complexity curve that isolates one important choice.

Students may use manual sweeps, random search, coarse-to-fine search, small structured grids, or targeted experiments. TAs should reward evidence-backed exploration rather than brute-force search. A smaller search can be acceptable when the student explains computational constraints and still provides a meaningful comparison. A search is weak if the tested values are arbitrary, too narrow, or selected based on the held-out test set.

Across all models, TAs should look for:

- leakage-safe preprocessing, especially scaling fit only on training data or training folds;
- training-only validation for tuning;
- learning curves with at least four training sizes unless justified;
- model-complexity curves with at least three reasonable values unless justified;
- final test metrics reported only after model selection;
- runtime reporting for fit and prediction;
- class-level error interpretation, especially for minority classes; and
- discussion that explains why models behaved differently.

A strong report should use these model families to tell a coherent supervised learning story. The models are not just boxes to run. They represent different inductive biases: threshold-based partitioning for Decision Trees, locality for kNN, margin-based classification and kernelized boundaries for SVMs, and capacity-driven gradient-based optimization for neural networks. The best submissions will connect those inductive biases to Covertypes's feature structure, class imbalance, curves, runtime, and observed error patterns.

Rubric

Total Points: 100 + 5 Extra Credit

TA Checks

Complete these checks before scoring.

Check 1: Readability

- Confirm that the submitted report PDF opens and is readable.
- Confirm that the submitted materials are sufficient to begin grading.

Action: If the report cannot be opened or evaluated, the assignment should receive a zero.

Check 2: Required Task

- Confirm that the report uses the Forest Cover Type / Covertype dataset.
- Confirm that the task is multiclass classification on **Cover_Type**.
- Confirm that the target classes are treated as classes 1 through 7, not collapsed into a binary target or treated as regression.

Action: If the report does not use Covertype or does not perform multiclass classification on **Cover_Type**, the assignment should receive a zero.

Check 3: Access

- Confirm that the report includes a read-only Overleaf link.
- Confirm that the Overleaf project uses the student's **Georgia Tech / Enterprise Overleaf account**, not a personal Overleaf account.
- Confirm that the reproducibility sheet includes a GitHub commit hash.
- Confirm that the repository uses the student's **GT Enterprise GitHub repository**, not a personal GitHub repository.
- Confirm that course staff can access the Overleaf project and GitHub repository without requesting edit permissions or personal-account invitations.

Action: If the required Overleaf project or GitHub repository is missing, inaccessible, or cannot be verified after escalation to the Head TA, the assignment may receive a zero for unverifiable provenance. If the artifacts are accessible but one access condition is incomplete or noncompliant, apply the scored Submission Access criteria and do not apply additional deductions.

Check 4: Page Limit

- Do not read or score content beyond the 8 page limit.

Action: If required content appears only after the page limit, treat that content as missing when applying the scored rubric.

Check 5: Draft Format

- Check whether the report reads like a formal technical paper rather than a checklist.
- Check for excessive bullet-only writing that replaces interpretation, especially in Results, Discussion, or Conclusion sections.
- Multiple instances of four or more consecutive bullets in analysis-heavy sections should be treated as excessive unless they are clearly part of a compact protocol, hyperparameter list, or reproducibility note.

Action: If the report contains excessive bullet-only writing that replaces interpretation, the report should not be scored and should be treated as a draft.

Rubric Points

For each of the following items, grade all-or-nothing. If an item is worth 2 points, award either 0 or 2 points. This helps maintain consistency across the grading staff.

Compliance (16 points)

Data Compliance (3 points)

- **(2 pts) Sample size.**
Award if: The report uses an approximately 20,000-instance working dataset and documents the sample size clearly.
Do not award if: The report uses the full dataset, an arbitrarily tiny subset, or a substantially different sample size without justification.
- **(1 pt) Source and target.**
Award if: The report identifies the data source and uses `Cover_Type` classes 1 through 7 consistently.
Do not award if: The data source is unclear or the label mapping is ambiguous.

Submission Access (4 points)

- **(2 pts) Enterprise Overleaf.**
Award if: The report includes a working read-only Overleaf link associated with the student's Georgia Tech / Enterprise Overleaf account.
Do not award if: The link is missing, broken, requires edit permissions, or appears to use a personal Overleaf account.
- **(2 pts) Enterprise GitHub.**
Award if: The report or reproducibility sheet includes a valid commit hash from the student's GT Enterprise GitHub repository, and the repository is accessible to course staff.
Do not award if: The hash is missing, invalid, inaccessible, or hosted only in a personal GitHub repository.

Reproducibility (3 points)

- **(2 pts) Run instructions.**
Award if: The student provides clear commands, environment requirements, data paths, and seeds sufficient to reproduce outputs on a standard Linux machine.
Do not award if: Instructions are missing or clearly insufficient to reproduce results.
- **(1 pt) Sampling procedure.**
Award if: The student provides the sampling method, sample size, random seed, and enough detail or code to recreate the sampled dataset.
Do not award if: Sampling is described vaguely or the sampled dataset cannot be recreated.

Figure Legibility (2 points)

- **(2 pts) Readable figures and tables.**

Award if: Figures, tables, axis labels, legends, captions, and text are readable at normal grading zoom, and the visual evidence can be evaluated without excessive zooming or guesswork.

Do not award if: Multiple figures or tables are too small, cropped, blurry, poorly labeled, or otherwise unreadable, making the visual evidence difficult to verify.

Attribution (4 points)

- **(2 pts) Outside sources.**

Award if: The report includes at least two relevant peer-reviewed sources beyond course materials and uses them to justify choices or interpret results.

Do not award if: Sources are missing, not peer-reviewed, irrelevant, only listed without use, or not connected to the report's methodology or interpretation. Textbooks, arXiv/preprint papers, blog posts, tutorials, documentation pages, lecture notes, and vendor materials may be useful background resources, but they do not satisfy the required peer-reviewed outside-source requirement.

- **(2 pts) AI usage statement.**

Award if: The report includes a clear AI/LLM usage statement.

Do not award if: The AI statement is missing, vague, does not identify tools used, does not describe how tools were used, or does not affirm that the student reviewed and understood all assisted content.

Study Design (14 points)

Goal: The report should justify the dataset sample, metrics, hypothesis, validation protocol, and leakage controls.

Extended EDA may appear in the reproducibility sheet, but the main report must include the evidence needed to justify sampling, metrics, hypotheses, preprocessing, and interpretation.

EDA Evidence (4 points)

- **(2 pts) Full-data EDA.**

Award if: The report includes full-dataset shape, feature types, missingness or absence of missingness, class counts, and class proportions.

Do not award if: Full-data EDA is absent, superficial, incorrect, or not connected to the task.

- **(2 pts) Sample EDA.**

Award if: The report gives class counts and class proportions for the approximately 20,000-instance sample.

Do not award if: Sample distribution is missing or unclear.

Sampling Design (4 points)

- **(3 pts) Sampling rationale.**

Award if: Stratification is used and justified, or non-stratified sampling is defended with evidence that all classes are preserved and the distribution is reasonable.

Do not award if: The report states that data was sampled randomly without evidence, rationale, or comparison to the full class distribution.

- **(1 pt) Ordering effects.**

Award if: The student uses a randomized or stratified procedure, or provides a compelling reason for a different choice.

Do not award if: The student appears to use `head(20000)` or the first rows without justification.

Metrics and Hypothesis (4 points)

- **(2 pts) Metric justification.**

Award if: The report explains why Accuracy, Macro-F1, balanced accuracy, confusion matrix, and per-class precision/recall/F1 are appropriate and why accuracy alone is insufficient.

Do not award if: Metrics are merely listed without explanation or only accuracy is meaningfully discussed.

- **(1 pt) Early hypothesis.**

Award if: At least one hypothesis is stated after EDA and before tuned results, predicts model behavior or trade-offs, and is tied to dataset properties and course theory.

Do not award if: The hypothesis is just a goal, is not testable, or is stated after results.

- **(1 pt) Hypothesis revisit.**

Award if: The report evaluates the hypothesis using metrics, learning curves, model-complexity curves, runtime, or error patterns.

Do not award if: The hypothesis is never revisited or is ignored in discussion.

Validation (2 points)

- **(1 pt) Test holdout.**

Award if: The test set is held out and used only once for final reporting.

Do not award if: Hyperparameters are chosen based on test performance or the final test set is repeatedly used for model selection.

- **(1 pt) Training-only validation and leakage controls.**

Award if: Hyperparameter tuning uses cross-validation or a fixed validation split on training data only, and preprocessing such as scaling is fit only on training data or training folds.

Do not award if: Validation is unclear, appears to include test data, or preprocessing appears to be fit before splitting.

Decision Trees (14 points)

- **(2 pts) Complexity control.**

Award if: The report identifies a control such as `max_depth`, `min_samples_leaf`, or `ccp_alpha`, and explains how it limits overfitting.

Do not award if: No pruning or regularization mechanism is used or explained.

- **(3 pts) Learning curve.**

Award if: The curve shows training and validation metric versus at least four training sizes and is used to discuss bias/variance.

Do not award if: The learning curve is missing, has fewer than four training sizes without justification, is mislabeled, or is not interpreted.

- **(3 pts) Complexity curve.**

Award if: The curve varies at least three reasonable values of a pruning or regularization parameter and supports the final model choice.

Do not award if: The curve is missing, uses too few values without justification, varies a parameter unrelated to complexity, or is not interpreted.

- **(2 pts) Final evaluation.**

Award if: Accuracy, Macro-F1, and balanced accuracy are reported, and confusion matrix or per-class precision/recall/F1 is included for this model or in the cross-model multiclass evaluation.

Do not award if: Required metrics are missing or error patterns are not discussed.

- **(2 pts) Runtime.**

Award if: Fit and predict time are reported with at least one trade-off.

Do not award if: Runtime is missing or only one timing value is reported without context.

- **(2 pts) Tuning quality.**

Award if: At least three reasonable hyperparameter values are tested, and the final choice is justified using validation or model-complexity evidence.

Do not award if: The final model appears arbitrary, only one setting is tested, or the tested values are too narrow to support the conclusion.

kNN (14 points)

- **(2 pts) Scaling and distance.**
Award if: Scaling is leakage-safe, and the distance metric and weighting choices are stated.
Do not award if: Scaling-sensitive preprocessing is missing, unclear, or fit before splitting.
- **(2 pts) k exploration.**
Award if: At least three distinct k values are tested and include meaningfully different neighborhood sizes, such as small, medium, and large values.
Do not award if: Only one k is tested, fewer than three values are tested without justification, or the tested values are too narrow to show bias/variance behavior.
- **(2 pts) Learning curve.**
Award if: The curve shows training and validation metric versus at least four training sizes and is interpreted.
Do not award if: The learning curve is missing, has fewer than four training sizes without justification, is mislabeled, or is not interpreted.
- **(2 pts) Complexity curve.**
Award if: The curve shows validation metric versus at least three distinct k values and supports the final model choice.
Do not award if: The curve is missing, uses too few k values without justification, or is not interpreted.
- **(3 pts) Final evaluation.**
Award if: Accuracy, Macro-F1, and balanced accuracy are reported, and confusion matrix or per-class precision/recall/F1 is included for this model or in the cross-model multiclass evaluation.
Do not award if: Required metrics are missing or error patterns are not discussed.
- **(2 pts) Runtime.**
Award if: Fit and predict time are reported, and prediction-time cost is discussed.
Do not award if: Runtime is missing or prediction-time cost is ignored.
- **(1 pt) Sensitivity discussion.**
Award if: The report discusses sensitivity to irrelevant features, high dimensionality, mixed continuous/binary features, or class imbalance.
Do not award if: No kNN-specific limitation is discussed.

SVM (14 points)

For computationally expensive SVM experiments, students may use a clearly justified validation strategy or reduced tuning grid, provided the comparison is reproducible and scientifically defensible.

- **(2 pts) Kernel comparison.**
Award if: Results from at least two kernels, such as linear and RBF, are reported and compared.
Do not award if: Only one kernel is used or the kernels are mentioned but not actually compared.
- **(2 pts) Scaling.**
Award if: Scaling is leakage-safe and applied consistently.
Do not award if: Scaling is missing, unclear, or fit before splitting.
- **(2 pts) Learning curve.**
Award if: The curve shows training and validation metric versus at least four training sizes and is interpreted.
Do not award if: The learning curve is missing, has fewer than four training sizes without justification, is mislabeled, or is not interpreted.
- **(2 pts) Complexity curve.**
Award if: The curve varies C and/or γ across at least three reasonable settings total, or provides a justified smaller grid for computational reasons.
Do not award if: The curve is missing, uses too few values without justification, varies an irrelevant parameter, or is not interpreted.

- **(3 pts) Final evaluation.**

Award if: Accuracy, Macro-F1, and balanced accuracy are reported, and confusion matrix or per-class precision/recall/F1 is included for this model or in the cross-model multiclass evaluation.

Do not award if: Required metrics are missing or error patterns are not discussed.

- **(1 pt) Runtime.**

Award if: Fit and predict time are reported and kernel trade-offs are discussed.

Do not award if: Runtime is missing or kernel runtime trade-offs are ignored.

- **(2 pts) Tuning quality.**

Award if: At least two kernels are evaluated, C is varied for each kernel where applicable, and γ is varied for RBF or another nonlinear kernel where relevant. A smaller grid is acceptable only with runtime justification.

Do not award if: The final model appears arbitrary, kernel choice is unsupported, or C/γ tuning is missing without justification.

Neural Networks (14 points)

If the student completes only one of the two required implementations, they should receive at most half credit for this section.

- **(3 pts) SGD setup.**

Award if: Optimizer is explicitly SGD, no disallowed optimizers are used, and learning rate, batch size, epochs, early stopping if used, and regularization are documented for both implementations.

Do not award if: Momentum, Nesterov, Adam, AdamW, Adagrad, RMSprop, or another adaptive optimizer is used, or the training setup is not documented.

- **(2 pts) Epoch curve.**

Award if: The report shows train and validation loss or metric over epochs for both the scikit-learn and PyTorch neural-network implementations and interprets optimization behavior.

Do not award if: Epoch evidence is missing, shown for only one implementation without justification, or not interpreted.

- **(2 pts) Learning curve.**

Award if: The curve shows training and validation metric versus at least four training sizes and is interpreted.

Do not award if: The learning curve is missing, has fewer than four training sizes without justification, is mislabeled, or is not interpreted.

- **(2 pts) Complexity curve.**

Award if: The curve varies at least three reasonable neural network settings, such as width, depth, learning rate, or regularization, and supports the final model choice.

Do not award if: The curve is missing, uses too few settings without justification, varies an irrelevant setting, or is not interpreted.

- **(2 pts) Final evaluation.**

Award if: Accuracy, Macro-F1, and balanced accuracy are reported, and confusion matrix or per-class precision/recall/F1 is included for the neural network models or in the cross-model multiclass evaluation.

Do not award if: Required metrics are missing or error patterns are not discussed.

- **(2 pts) Implementation comparison.**

Award if: The report compares scikit-learn and PyTorch in terms of optimization behavior, stability, runtime, tunability, or implementation differences under SGD-only constraints.

Do not award if: Both implementations are reported but not meaningfully compared.

- **(1 pt) Runtime and tuning.**

Award if: At least three reasonable NN settings are tested, and fit/predict time or epoch-level training cost is reported and interpreted.

Do not award if: The final model appears arbitrary, too few settings are tested without justification, or runtime/training cost is missing.

Discussion (14 points)

Goal: The report should move beyond presenting algorithm results and explain what the full set of experiments reveals about the dataset, the algorithms, and the modeling trade-offs.

The discussion should be written in paragraph form. It should not be a second results table or a bullet-only recap of scores.

- **(3 pts) Cross-model synthesis.**

Award if: The discussion directly compares at least three of the required algorithm families and explains why their behavior differed using evidence from final metrics, learning curves, model-complexity curves, runtime, and/or error patterns. A strong discussion should connect observed behavior to course concepts such as bias-variance, locality, margin-based classification, scaling sensitivity, model capacity, or regularization.

Do not award if: The report presents each model separately but does not compare them, only states which model scored highest, or gives unsupported claims such as “SVM performed best because it is powerful.”

- **(3 pts) Performance and efficiency trade-offs.**

Award if: The discussion explains trade-offs between predictive performance and computational cost. This should include at least two concrete comparisons, such as accuracy/Macro-F1 versus fit time, prediction time for kNN, SVM kernel cost, tree interpretability versus performance, or neural-network training stability versus runtime.

Do not award if: The discussion reports runtime and metrics separately but does not explain trade-offs, ignores computational cost, or declares a winner using only one metric.

- **(3 pts) Class-level behavior.**

Award if: The discussion analyzes multiclass performance rather than only aggregate scores. It should identify which cover types were easiest or hardest to classify, discuss minority-class behavior, and use the confusion matrix or per-class precision/recall/F1 to explain at least one important misclassification pattern.

Do not award if: The discussion ignores class imbalance, only reports overall accuracy, does not mention minority classes, or includes a confusion matrix without interpreting class-level errors.

- **(3 pts) Curve-based interpretation.**

Award if: The discussion uses learning curves and model-complexity curves to explain model behavior. It should make at least two specific claims about underfitting, overfitting, variance, bias, saturation with more data, or hyperparameter sensitivity, and those claims should be tied to the plotted curves.

Do not award if: Curves are included but only described visually, such as “the line went up,” or the discussion does not use the curves to support claims about bias, variance, underfitting, overfitting, or tuning behavior.

- **(2 pts) Limitations and improvements.**

Award if: The discussion identifies at least one concrete limitation of the study and at least one evidence-based improvement or follow-up experiment. Acceptable limitations include the 20,000-instance sample, minority-class instability, limited SVM grid size due to runtime, SGD-only neural-network constraints, class imbalance, or restricted feature engineering. The improvement must follow from the evidence, such as testing class weighting after poor minority recall, expanding the SVM grid after sensitivity to C , increasing minority-class representation, or revisiting NN learning rates after unstable epoch curves.

Do not award if: Limitations are missing, generic, or disconnected from the results. Improvements such as “use more data,” “try more models,” or “tune better” do not receive credit unless tied to specific evidence from the report.

Extra Credit: Activation Study

Up to 5 points. Extra credit must follow the SGD-only rule. If a disallowed optimizer is used, no extra credit should be awarded.

- **(2 pts) Activation curves.**

Award if: Epoch-wise validation curves are shown for all four activations using the same training budget.

Do not award if: Curves are missing, not comparable, or use different training protocols.

- **(2 pts) Comparison table.**

Award if: A final comparison table reports metrics, runtime, and convergence notes.

Do not award if: The comparison table is missing or incomplete.

- **(1 pt) Interpretation.**

Award if: The report explains convergence stability and trade-offs under SGD.

Do not award if: The activation study only reports numbers without interpretation.

Version Control

- v1.0 – 05/19/2026 – TJL refactored SL rubric for Summer 2026 Covertypes assignment.

Rubric written by Theodore LaGrow.